

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

- **Date:** 18 June 2026
- **Project Name:** Shopez

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	• Customer registration and secure login. • Role-based login routes for Admin vs Customer.
FR-2	Product Catalog	• Render lists of items with prices and stock levels. • Fallback catalog loading from JSON files if database is disconnected.
FR-3	Shopping Cart	• Add/Remove items from cart. • Dynamic price calculation including taxes and shipping.
FR-4	Checkout System	• Credit card processing mockup with animated visual graphic. • Order placement and inventory stock deduction on purchase.
FR-5	Admin Control Panel	• Insights tab displaying total sales, user list, and inventory status. • Visibility fix: black text for easy contrast reading.

Non-functional Requirements

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clean, responsive light-themed layout styled with Vanilla CSS. UI is intuitive.
NFR-2	Security	Hashed user passwords in database. Credentials are not hardcoded or shown on screen.
NFR-3	Reliability	Application degrades gracefully by loading from local JSON database files when MongoDB fails.
NFR-4	Performance	Sub-second API response times for cart and catalog endpoints.
NFR-5	Availability	Up-time maximized on cloud deployment (Render/Vercel) with reliable fallback methods.

FR No.	Non-Functional Requirement	Description
NFR-6	Scalability	Separation of client (Vite/React) and server (Node/Express) allows independent horizontal scaling.